David Neto

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Education

Doctor of Philosophy - Physics — Dec 2025
Dissertation: Use of Active Target Detectors in Nuclear Astrophysics: Nucleosynthesis of Fluorine and Fusion of Neon
University of Illinois Chicago, Chicago, IL

Bachelor of the Arts (cum laude) - Physics, Mathematics — May 2011
Senior Project: A Study of Neutralino Two Decays Using SUSY-HIT
Rhode Island College, Providence, RI

Research Experience

Facility for Rare Isotope Beams — East Lansing, MI
- *CeNAM - Research Associate* — Dec '25 – Present

Argonne National Laboratory — Lemont, IL
- *Research Aide Technical - PhD* — Nov '23 – Jul '25

- Performed the first measurement of the ^{34}Ar(α,p)^{37}K cross section at X-ray burst energies using the Argonne MUSIC detector.
- Adapted the machine learning techniques and methods developed for fusion analysis for use in (α, p) and (α, n) MUSIC experiments, exploring future improvements and applications for the machine learning pipeline.
- Mentored an undergraduate student in stellar astrophysics and the simulation of Type I X-ray bursts with the 1D stellar evolution code MESA.

Argonne National Laboratory — Lemont, IL
- *DOE SCGSR Graduate Student* — Nov '22 – Nov '23

- Documented a purely Python-based analysis framework for MUSIC data, openly available on GitHub with detailed examples.
- Developed and tested a machine learning pipeline for analysis of fusion reactions measured with the Argonne MUSIC detector.
- Carried out an experiment to measure fusion of neon reactions ^{20,22,24}Ne + ^{20,22}Ne with the Argonne MUSIC detector.
- Built and trained a convolutional neural network classifier for fusion event identification.

University of Illinois Chicago — Chicago, IL
- *Bubble Collaboration - Graduate Student* — May '18 – Nov '22

- Joint ANL and JLab bubble chamber experiment to measure reaction rates of astrophysical importance.
- Assisted with data collection for the 2018 experimental run at Jefferson Lab with the goal to measure the cross section of the ^{19}F(γ,α)^{15}N reaction using a liquid C_3F_8 target and a bremsstrahlung beam.
- Responsible for data analysis, determining detector efficiency over the different experimental runs, and computing the experimental yield.
- Performed Geant4 simulations on the JLab compute farm to obtain γ-ray profiles for the different beam energies, determined γ-ray beam size in the target, and estimate potential backgrounds from (γ,n) reactions.

Oklahoma State University — Stillwater, OK
- *Research Assistant - Dept. of Physics* — May '15 – May '17

- Research work in the Infrared Structural Biology Lab of Dr.Aihua Xie concerning the use of FTIR spectroscopy to determine the secondary structure of proteins.
- Took measurements with FTIR spectrometer using both transmission and ATR type cells, performed analysis of spectra and fitting of spectroscopic data with NLLS fitter.
- Experience with both Cyro-cooled MCT and DLaTGS infrared detectors.
- Mentored undergraduates in the use of the FTIR system, how to load/measure samples and analyze spectra.

- *Research Assistant - Dept. of Physics* May '14 – Sep '14

- Research work for the ATLAS collaboration of CERN under Dr. Joesph Haley.
- Studied pruning algorithms for the boosted boson group using 7 and 8 TeV Monte Carlo and Run 1 experimental data using ROOT.
- Computed ROC curves, Mass/Energy response plots and correlation plots.
- Helped to determine ideal pruning parameters for both C/A and k_t re-clustering algorithms over jet radius, R_{cut} and Z_{cut} parameters for a given transverse momentum range.

Rhode Island College — Providence, RI
- *Student Researcher - Dept. of Physical Science* Sep '10 – May '11

- Research project studying Neutralino decay in mSUGRA under the supervision of Dr. Brandon Murakami.
- Involved directed study of QFT, at the level of Part I of Peskin & Schroeder, and an introduction to N=1 SUSY.
- Computed decay widths and branching ratios using SUSY spectrum tools SOFTSUSY and Susy-HIT.
- Introduction to simple e^+e^- collider simulations in PYTHIA and computation of Feynman diagrams using CompHEP.

Awards

APS DNP Dissertation Award Oct'26
"For outstanding contributions to active-target programs and nuclear-fusion studies in experimental nuclear astrophysics, through innovative detector research and data analysis as well as advanced machine-learning applications."

DOE SCGSR Award Nov '22

UIC Graduate Student Council Travel Award May '19

2nd Place Oral Presentation, BMBGSA Symposium (Oklahoma State) Sep '16
"Formation of Amino Acids via Astrophysical Processes"

Paul Westhaus Scholarship (Oklahoma State) May '16

2nd Place Oral Presentation, BMBGSA Symposium (Oklahoma State) Sep '15
"Determining the Secondary Structure of Proteins Using FTIR Spectroscopy"

Rhode Island College Departmental Physics Award May '11

Ronald J. Boruch Physical Sciences Award (Rhode Island College) May '11

Papers

E. Lopez-Saavedra, M. L. Avila, W.-J. Ong, P. Mohr, S. Ahn, H. Arora, L. Balliet, K. Bhatt, S. M. Cha, K. A. Chipps, J. Dopfer, I. A. Tolstukhin, R. Jain, M. J. Kim, K. Kolos, F. Montes, **D. Neto**, S. D. Pain, J. Pereira, J. S. Randhawa, L. J. Sun, C. Ugalde, L. Wagner (2026) *Detailed Study of the $^{59}Cu(p,\alpha)^{56}Ni$ Reaction and Constraints on Its Astrophysical Reaction Rate.* arXiv:2601.22262 [nucl-ex]

E. Lopez-Saavedra, M. L. Avila, W.-J. Ong, P. Mohr, S. Ahn, H. Arora, L. Balliet, K. Bhatt, S. M. Cha, K. A. Chipps, J. Dopfer, I. A. Tolstukhin, R. Jain, M. J. Kim, K. Kolos, F. Montes, **D. Neto**, S. D. Pain, J. Pereira, J. S. Randhawa, L. J. Sun, C. Ugalde, L. Wagner (2026) *Direct Measurement of the $^{59}Cu(p,\alpha)^{56}Ni$ Excitation Function to Constrain the Ni–Cu Cycle Strength and Its Impact on Explosive Nucleosynthesis.* arXiv:2601.22262 [nucl-ex]

S. Bhattacharya, M. Paul, R.N. Sahoo, R. Purtschert, H.F.R. Hoffmann, M. Pichotta, K. Zuber, D. Bemmerer, T. Döring, R. Schwengner, M.L. Avila, E. Lopez-Saavedra, J. C. Dickerson, C. Fougères, J. McLain, R. C. Pardo, K. E. Rehm, R. Scott, I. Tolstukhin, R. Vondrasek, T. Bailey, L. Callahan, A. M. Clark, P. Collon, Y. Kashiv, A. Nelson, D. Robertson, **D. Neto**, C. Ugalde, M. Tessler, S. Vaintraub (2025) *First experimental determination of the 40Ar(n,2n)39Ar reaction cross section and 39Ar production in Earth's atmosphere.* Geochim. Cosmochim. Acta 415, 196

M. L. Avila, K. Auranen, A. D. Ayangeakaa, K. Bhatt, J. Chen, C. Fougères, C. R. Hoffman, H. Jayatissa, C. L. Jiang, B. P. Kay, E. Lopez-Saavedra, **D. Neto**, W.-J. Ong, K. E. Rehm, D. Santiago-Gonzalez, C. Ugalde, J. Wu (2025). *Understanding X-ray bursts via direct measurements of (α,p) reactions.* Nucl. Phys.

A 1063. 123212

C. Fougères, M. L. Avila, A. Psaltis, M. Anastasiou, S. Bae, L. Balliet, K. Bhatt, L. Dienis, H. Jayatissa, V. Karayonchev, P. Mohr, F. Montes, **D. Neto**, F. de Oliveira Santos, W.-J. Ong, K. E. Rehm, W. Reviol, D. Santiago-Gonzalez, N. Sensharma, R. S. Sidhu, I. A. Tolstukhin (2025). *First Measurement of 87Rb(α,xn) Cross Sections at Weak r-process Energies in Supernova ν-driven Ejecta to Investigate Elemental Abundances in Low-metallicity Stars.* Astrophys. J. 983, 142

C. Fougères, M. L. Avila, H. Jayatissa, D. Santiago-Gonzalez, K. Brandenburg, Z. Meisel, P. Mohr, F. Montes, C. Műller-Gatermann, **D. Neto**, W.-J. Ong, J. Pereira, K. E. Rehm, T. L. Tang, I. A. Tolstukhin, L. Varriano, G. Wilson, J. Wu (2024). *Direct cross-section measurement of the weak r-process 88Sr(α,n)91Zr reaction in ν-driven winds of core collapse supernovae.* Phys. Rev. C 109, 065805

H. Jayatissa, M. L. Avila, K. E. Rehm, P. Mohr, Z. Meisel, J. Chen, C. R. Hoffman, J. Liang, C. Müller-Gatermann, **D. Neto**, W.-J. Ong, A. Psaltis, D. Santiago-Gonzalez, T. L. Tang, C. Ugalde, G. Wilson (2023). *Study of the ^{22}Mg waiting point relevant for X-ray burst nucleosynthesis via the ^{22}Mg(α,p)^{25}Al reaction.* Phys. Rev. Lett. 131, 112701

D. Neto, K. Bailey, J. F. Benesch, B. Cade, B. DiGiovine, A. Freyberger, J. M. Grames, A. Hofler, R. J. Holt, R. Kazimi, D. Meekins, M. McCaughan, D. Moser, T. O'Connor, M. Poelker, K. E. Rehm, S. Riordan, R. Suleiman, R. Talwar, C. Ugalde (2023). *Measuring the cross section of the ^{15}N$(\alpha,\gamma)^{19}$F reaction using a single-fluid bubble chamber.* Phys. Rev. C 107, 035801

L. Asquith, W. Bhimiji, P. Bokan, N. Bousson, T. Bristow, R. Camacho, M. Campanelli, R. Chislett, B. Cooper, C. Delitzsch, F. Dias, E. Gozani, J. Haley, S. Hsu, E. Kajomovitz, D. Kar, F. De Lorenzi, S. Meehan, B. Nachman, **D. Neto**, Y. Rozen, J. Veatch, L. Zivkovic (2014). *Performance of Boosted W Boson Identification with the ATLAS Detector.* ATL-COM-PHYS-2014-1450. CERN.

Invited Talks

"From Bubbles to MUSIC: Experiments with Active Target Detectors" — Oct '26
APS Fall DNP Meeting, Philadelphia, PA

"Measuring the cross section of the ^{15}N(α,γ)^{19}F reaction using a single-fluid bubble chamber" — Jul '24
CAARI-SNEAP, Fort Worth, TX

Presentations

"Updating the REACLIB thermonuclear reaction rate library" — Jul '26
Yale/CeNAM Nuclear Asteroseismology Workshop, New Haven, CT

"The fusion of neon measured with the MUSIC detector at Argonne: An analysis using machine learning." — Mar '25
APS Joint March Meeting and April Meeting: Global Physics Summit 2025, Anaheim, CA

"An R-Matrix analysis of the 5337 keV resonance in 19F." — Apr '24
APS Fall Meeting, Sacramento, CA

"Measuring the cross section of the 15N(a,g)19F reaction using a single-fluid bubble chamber." — Oct '21
APS DNP Fall Meeting (Virtual)

"A single fluid bubble chamber for measuring nuclear reaction rates of astrophysical importance" — Apr '19
APS April Meeting, Denver, CO

"Formation of Amino Acids via Astrophysical Processes" — Sep '16
BMBGSA Symposium, Oklahoma State University, Stillwater, OK

"Determining the Secondary Structure of Proteins Using FTIR Spectroscopy" — Sep '15
BMBGSA Symposium, Oklahoma State University, Stillwater, OK

Workshops and Summer Schools

Yale/CeNAM Nuclear Asteroseismology Workshop — Jul '26
Yale, New Haven, CT

.Astronomy 12 — Oct '23
Center for Computational Astrophysics, New York, NY

Exotic Beam Summer School — Jun '22
University of Notre Dame, South Bend, IN

MESA Summer School — Aug '21
(Virtual)

ECT* Talent School — Jul '21
Machine Learning Applied to Nuclear Physics (Virtual)

.Astronomy 11 — Oct '19
University of Toronto, Toronto, Canada

AIP ComSciCon — Sep '19
College Park, MD

Chicago ComSciCon — Aug '18
Chicago, IL

Teaching Experience

University of Illinois Chicago — Chicago, IL
- *Assistant to the Undergraduate Lab Coordinator* — Sep '19 - Dec '22

- Setup of lab stations and testing of new lab equipment.
- Helped with conversion of in-person to online labs due to COVID restrictions
- Updated and redesigned "Intro to Astronomy" Labs

University of Illinois Chicago — Chicago, IL
- *Teaching Assistant* — Sep '17 - May '19

- Responsible for 2-3 sections of algebra based Physics II (E&M) labs.
- Responsible for 1 section of Intro to Astronomy labs.
- Weekly office hours as a physics tutor in the Math and Science Learning Center.

Oklahoma State University — Stillwater, OK
- *Teaching Assistant* — Sep '13 – May '14, Jan '16 - May '17

- Taught 4 sections of calculus based General Physics 1 lab, approximately 80 students per semester.
- Prepared pre-lab presentations in addition to grading lab reports and holding study sessions.

Rhode Island College — Providence, RI
- *Department Physics Tutor* — Sep '09 – May '11

- Helped students in both calculus and algebra based Physics I and II.
- Held exam study sessions and prepared custom study guides for students

Posters

"New REACLIB" — May '26
Frontiers in Nuclear Astrophysics Meeting 2026, College Station, TX

"Measuring the $^{15}N(\alpha,\gamma)^{19}F$ reaction using a bubble chamber" — Jun '19
AAS Summer Meeting, St.Louis, MO

"Improved methods for determination of protein secondary structure from FTIR spectroscopy." — Mar '16
APS March Meeting, Baltimore, MD

Skills

Programming Languages: Python, C++
Software: MESA, Mathematica, ROOT, Geant4, LaTeX
Packages & Libraries: NumPy, SciPy, Matplotlib, scikit-learn, Keras
Languages: English (native), Italian (conversational)

Membership

American Physical Society, American Astronomical Society

Work Experience

Property Manager, Cargill Farms, Cumberland, RI — May '11 - June '13
- Maintenance of buildings on property involving various electrical, plumbing and carpentry tasks.
- Upkeep and repair of farm equipment.
- Landscaping duties, grass cutting and leaf/debris removal.